

Expander Graphs

Guo Haoyang

March 2026

Contents

1	Basic properties on expander graphs	2
1.1	Basic properities	2
1.1.1	Adjacency operators	2
1.1.2	Expanding perperty	2
1.2	Characterisation of an expander graph	5
2	Constructing expander graphs	9
2.1	Cayley graphs	9
2.2	Schreier graphs	9
2.3	Kazhdan's property (T)	9
2.4	Zigzag product constructions	10
3	Applications of expander graphs:	10
4	Applications of expander graphs: to certain sieving problems in analytic number theory	10

1 Basic properties on expander graphs

1.1 Basic properties

1.1.1 Adjacency operators

Let $\Gamma = (V, E)$ be a connected, undirected k -regular graph with n vertices and without loops (so the diagonal entries of the adjacency matrix are 0). We consider the Hilbert space of real-valued functions on the vertex set, denoted as $\ell^2(V)$, equipped with the standard inner product $\langle f, g \rangle = \sum_{v \in V} f(v)g(v)$. We define the adjacency operator $A : \ell^2(V) \rightarrow \ell^2(V)$ by its action on a function f :

$$(Af)(x) = \sum_{y \in N(x)} f(y)$$

where $y \sim x$ denotes that y is adjacent to x .

Because the graph is undirected, the operator A is self-adjoint (symmetric), meaning $\langle Af, g \rangle = \langle f, Ag \rangle$ for all $f, g \in \ell^2(V)$. By the Spectral Theorem for finite-dimensional Hilbert spaces, a self-adjoint operator is diagonalizable and possesses a purely real spectrum. Thus, $\ell^2(V)$ admits an orthonormal basis of eigenfunctions. We can order its n real eigenvalues as:

$$\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_n \quad (1)$$

Since Γ is a k -regular graph, the constant function $\mathbf{1}$ is an eigenfunction with eigenvalue k because for every $v \in V$,

$$A\mathbf{1}(v) = \sum_{u \in N(v)} \mathbf{1}(u) = \sum_{u \in N(v)} 1 = k$$

The entire spectrum is within $[-k, k]$. Suppose λ is an arbitrary eigenvalue. For any $f \in \ell^2(V)$, f must attain its absolute maximum at some $x_0 \in V$ because of the finiteness of V . Hence, for any $y \in V$ and any $f \in \ell^2(V)$,

$$\lambda f(x_0) = (Af)(x_0) = \sum_{y \in N(x_0)} f(y)$$

Taking the absolute value,

$$|\lambda| \cdot |f(x_0)| = \left| \sum_{y \in N(x_0)} f(y) \right| \leq \sum_{y \in N(x_0)} |f(y)| \leq \sum_{y \sim x_0} |f(x_0)| = k \cdot |f(x_0)|$$

f could not be always zero, therefore $|\lambda| \leq k$. This implies that $\lambda_1 = k$. Furthermore, $\lambda_n = -k$ if and only if Γ is a bipartite graph (i.e., a graph whose vertex set can be partitioned into two disjoint sets such that every edge connects a vertex in one set to a vertex in the other).

1.1.2 Expanding property

Definition 1.1: Expander graphs

Let $\varepsilon > 0$ and $k \geq 1$. A finite k -regular graph is said to be a **(one-sided) ε -expander** if one has

$$\lambda_2 \leq (1 - \varepsilon)k$$

and a **two-sided ε -expander** if one also has

$$\lambda_n \geq -(1 - \varepsilon)k$$

A sequence $G_i = (V_i, E_i)$ of finite k -regular graphs is said to be a one-sided (resp. two-sided) **expander family** if there is an $\varepsilon > 0$ such that G_i is a one-sided (resp. two-sided) ε -expander for all sufficiently large i .

(The notation used in this definition follows T. Tao's blog)

The definition gives us no intuition what an expander graph should look like. In the next proposition, we shall detect some necessary condition for an expander graph.

Proposition 1.1: Necessary condition of being an expander graph

Let Γ be a connected k -regular graph. Then $\lambda_1 = k$ is simple (namely, $\lambda_1 > \lambda_2$). If Γ is also bipartite, then $\lambda_n = -k$ is also simple.

Remark If Γ is not connected and has l connected components $G_1, G_2, \dots, G_l$, then the multiplicity of the eigenvalue k is exactly l . The spectrum of Γ is a disjoint union of the spectra of G_i , so the spectral gap of Γ is the minimum of the spectral gaps for $G_1, \dots, G_l$.

The first condition simply says that for a regular graph, the 'least' property of being an expander graph is connectivity.

Proof: (1) Suppose that Γ is connected. Once the multiplicity of λ_1 is 1, $\lambda_1 \neq \lambda_2$. Suppose that f is the eigenvector corresponds to the eigenvalue k , then $\forall v \in V, Af(v) = kf(v)$. Meanwhile, $Af(v) = \sum_{u \in N(v)} f(u)$. Equating them, we have

$$kf(v) = \sum_{u \in N(v)} f(u)$$

Since f is a function on a finite set V , f attains its maximum M at some x_0 , i.e. $f(x_0) = M$. Then,

$$kM = k \cdot f(x_0) = \sum_{u \in N(x_0)} f(u) \leq kM$$

The only possible case for $f(u)$ with $u \in N(x_0)$ is: $f(u) = M$. The connectivity of Γ allows the spreading of M so f is a constant function, which means that the eigensubspace V_{λ_1} has dimension 1. This is equal to the multiplicity of λ_1 . So, k only appears once.

(2) When $\lambda_n = -k$, this $-k$ corresponds to some eigenvector $f \in l^2(V)$. Applying A to f , we obtain

$$Af = -kf$$

Since f is on a finite set V . It suffices to take f as a real valued function. Then, the absolute value of f must have the maximum and minimum, say $N \leq |f(v)| \leq M$ for some M and N . Suppose that f attains the maximum at x_0 and minimum at x_1 .

Evaluating f at x_0 , one can equate $Af(x_0) = -kf(x_0) = -kM$ and $Af(x_0) = \sum_{u \in N(x_0)} f(u)$,

$$-kM = \sum_{u \in N(x_0)} f(u)$$

From the value of $f(u)$, $f(u) \geq -M$. Therefore, $-kM = \sum_{u \in N(x_0)} f(u) \geq k \cdot (-M)$. The only possibility for $f(u)$ is $f(u) = -M$ for each $u \in N(x_0)$.

Suppose that when Γ contains a non-empty bipartite graph as a connected component, denoted C . This component C attains a bipartition X and Y . Construct a function $g : V \rightarrow \mathbb{C}$ ($g \in l^2(V)$) by

$$g(v) = \begin{cases} 1 & v \in X \\ -1 & v \in Y \\ 0 & v \notin C \end{cases}$$

Applying the adjoint operator A on this g , we have the three cases:

(i) $\forall v \in X$,

$$Ag(v) = \sum_{u \in N(v)} g(u) \stackrel{\dagger}{=} \sum_{u \in N(v)} -1 = -k = -k \cdot g(v)$$

$\dagger$: In the bipartite subgraph C , all neighbourhood of $v \in X$ are in Y . Hence, $g(u) = -1$.

(ii) $\forall v \in Y$,

$$Ag(v) = \sum_{u \in N(v)} g(u) = k = -k \cdot g(v)$$

because $v \in Y$ and $N(v) \subseteq X$.

(iii) $\forall v \in C$,

$$Ag(v) = \sum_{u \in N(v)} g(u) = 0 = -k \cdot g(v)$$

which is because that neighbourhoods of $v \in C$ are still in C .

So, under each case, the equality

$$A \cdot g = -k \cdot g$$

holds. $-k$ is an eigenvalue of A . By the discussion before (1), the eigenvalues of A are in $[-k, k]$. So, it must be $\lambda_n = -k$. $\square$

Using this proposition, we are able to give some examples and non-examples of expander graphs now:

- Consider the graph whose vertex set is $\mathbb{Z}/n\mathbb{Z}$ and edges are those $([i], [i+1])$. This is a 2-regular graph but not an expander graph.
- Complete graphs K_n are all expander graphs.

As a complete graph, it owns too much connectivity than being an expander graph. While, as $\mathbb{Z}/n\mathbb{Z}$, it has too less connectivity than being an expander graph.

The following are in [Big93].

Proposition 1.2: Trace formulae

Let Γ be a k -regular graph with $|V| = n$ for some $n > k \geq 1$. Then,

- (1) $\text{Tr}(A) = \sum_{i=1}^n \lambda_i = 0$
- (2) $\text{Tr}(A^2) = \sum_{i=1}^n \lambda_i^2 = nk$.
- (3) $\max(|\lambda_2|, |\lambda_n|) \geq \sqrt{k} - o_{n \rightarrow \infty; k}(1)$, where $o_{n \rightarrow \infty; k}(1)$ denoted a quantity that goes to zero as $n \rightarrow \infty$ for fixed k .

Proof: (1) The sum of the eigenvalues is exactly the trace of the operator A :

$$\sum_{i=1}^n \lambda_i = \text{Tr}(A) = \sum_{v \in V} \langle \delta_v, A\delta_v \rangle$$

By the definition of the adjacency operator, $(A\delta_v)(x)$ evaluates to 1 if x is a neighbor of v , and 0 otherwise. Since the graph is simple (lacking self-loops), a vertex is never a neighbor of itself. Therefore, $(A\delta_v)(v) = 0$. Consequently, the inner product $\langle \delta_v, A\delta_v \rangle = \delta_v(v) \cdot (A\delta_v)(v) = 1 \cdot 0 = 0$. Summing over all vertices yields $\text{Tr}(A) = 0$.

(2) Similarly, the sum of the squared eigenvalues corresponds to the trace of the squared operator A^2 :

$$\sum_{i=1}^n \lambda_i^2 = \text{Tr}(A^2) = \sum_{v \in V} \langle \delta_v, A^2\delta_v \rangle$$

Because A is a self-adjoint operator ($A = A^*$), we can shift one A to the other side of the inner product:

$$\langle \delta_v, A^2\delta_v \rangle = \langle A^*\delta_v, A\delta_v \rangle = \langle A\delta_v, A\delta_v \rangle = \|A\delta_v\|^2$$

The function $A\delta_v$ acts as the indicator function for the neighborhood of v . It takes the value 1 on exactly k vertices (since the graph is k -regular) and 0 elsewhere. The squared ℓ^2 -norm of such a function is simply the count of those vertices, which is k . Thus, for every vertex v , $\|A\delta_v\|^2 = k$. Summing this over all n vertices gives $\text{Tr}(A^2) = nk$. $\square$

Remark The intuition of A lies in its matrix representations. The entry at i -th column and j -th row is 1 if there is an edge between the i -th vertex and the j -th vertex. Otherwise, it is zero. The entries on the diagonal are all zero because loops are not allowed (there is no edge from a vertex to itself). So, the trace of A is 0.

Remark As a general property for graphs, the sum of the q -th powers of the eigenvalues equals the trace of A^q , which has a precise combinatorial meaning. Specifically,

$$\sum_{i=1}^n \lambda_i^q = \text{Tr}(A^q) = \text{number of closed paths of length } q \text{ on } \Gamma$$

This is a standard result in algebraic graph theory (see, for example, Biggs, *Algebraic Graph Theory*).

Corollary 1.1

For a complete graph K_n , the spectra of K_n , λ_i (except the largest one λ_1), are all equal to -1 :

$$\lambda_2 = \lambda_3 = \dots = \lambda_n = -1$$

For a graph Γ and its complement Γ^c , their spectra satisfies the equality

Proof:

□

1.2 Characterisation of an expander graph

To understand why being merely "connected" is insufficient, let us examine the cycle graph $\mathbb{Z}/n\mathbb{Z}$. Intuitively, this graph lacks robust connectivity. If we want to travel from one vertex to its antipodal vertex—for instance, from vertex 0 to vertex $\lfloor n/2 \rfloor$ —we are forced to traverse half the cycle. As the total number of vertices $n \rightarrow \infty$, this path length diverges to infinity, meaning distant vertices cannot be reached directly. We can view this from the dynamic perspective of a random walk. In a random walk, we care about the number of steps required for a walker's probability distribution to diffuse evenly across the entire graph (a concept known as mixing time). On a cycle graph, this diffusion process is exceedingly slow, as the walker is restricted to moving only to immediate local neighbors step by step. It is precisely because of this sluggish diffusion that the second largest eigenvalue λ_2 asymptotically approaches the largest eigenvalue λ_1 . Consequently, the spectral gap vanishes to zero, making the cycle graph a prime example of a poor expander.

Therefore, we must measure not only whether a graph is connected, but the quality of that connectivity. To evaluate this, inspecting the connections between individual pairs of vertices is inadequate; instead, we must examine how any local region S (a subset of vertices) connects to the outside of that region. Imagine partitioning the vertex set into two disjoint halves S and $V \setminus S$. To completely separate these two parts into disconnected components, we must cut certain edges. The set of edges bridging the two subsets is known as the edge boundary, denoted ∂S . The size of this boundary dictates the strength of the connection between the two parts. If two parts are connected by only a minimal number of edges, they form a "bottleneck," meaning the connectivity between them is extremely fragile and easily broken.

Crucially, examining just a single subset and its exterior is insufficient to evaluate the overall connectivity of the graph. To accurately pinpoint the graph's most vulnerable bottleneck, we must scan through every possible local region and measure its edge boundary.

The core philosophy behind Cheeger's approach is that the absence of such bottlenecks characterizes a good expander. In a highly expansive graph, no matter how we partition the vertex set, there will always be a substantial number of cross-edges connecting the two halves. To formalize this geometric intuition rigorously, we arrive at the following definition:

Definition 1.2: Cheeger constant

$$h(G) = \min_{S \subset V, |S| \leq \frac{|V|}{2}} \frac{|\partial S|}{|S|}$$

Remark The intuition of the ratio $\frac{|\partial S|}{|S|}$ is for a local region with size $|S|$, the average number of paths towards the outside of S .

Taking min implies the examination of all possible set S . The bottleneck of being 'strong connectivity' (or expander) depends on the part with the most vulnerable connection towards its outside.

The restriction of $|S|$ is because the following reasons:

(i) Once the vertex set V is partitioned into two parts S and $V \setminus S$ (or S^c). They share the same boundary, namely $\partial S = \partial S^c$. At least one of $|S|$ and $|S^c|$ is less than $\frac{|V|}{2}$.

(ii) If there were no restriction $|S| \leq \frac{|V|}{2}$, then $|S|$ would be very large (for example, almost all vertices of V). In this case, the ratio approaches to almost zero but the graph could have relatively robust connectivity (satisfy the expander condition). So, the ratio would not reveal the expander of the graph. To make the 'bottleneck' being reflected explicitly or not covered, the restriction $|S| \leq \frac{|V|}{2}$ makes us examine ∂S from the smaller part.

Proposition 1.3: Weak discrete Cheeger inequality

Let $k \geq 1$ and let $\{\Gamma_n\}$ be a family of finite k -regular graphs.

(1) $\{\Gamma_n\}$ forms a one-sided expander family.

(2) $\exists C > 0, \exists N, \forall n > N, h(\Gamma_n) \geq C$.

Let D be a degree operator, which means that for any function $f \in \ell^2(V)$,

$$(Df)(v) = \deg(v)f(v)$$

For an operator $d : \ell^2(V) \rightarrow \ell^2(E)$ such that $(df)(e) := f(e^+) - f(e^-)$ for all $f \in \ell^2(V)$. There is an adjoint operator, $d^* : \ell^2(E) \rightarrow \ell^2(V)$ for this d such that $d^*F(v) = \sum_{e \in E^-(\{v\})} F(e) - \sum_{e \in E^+(\{v\})} F(e)$ for all $F \in \ell^2(E)$. Applying d^*d to any function $f \in \ell^2(V)$, we have

$$\begin{aligned} (d^*d)f(v) &= \sum_{e \in E^+(\{v\})} (df)(e) - \sum_{e \in E^-(\{v\})} (df)(e) \\ &= \sum_{e \in E^-(\{v\})} (f(e^+) - f(e^-)) - \sum_{e \in E^+(\{v\})} (f(e^+) - f(e^-)) \\ &= \left(\sum_{e \in E^-(\{v\})} f(e^+) + \sum_{e \in E^+(\{v\})} f(e^-) \right) - \left(\sum_{e \in E^-(\{v\})} f(e^-) + \sum_{e \in E^+(\{v\})} f(e^+) \right) \\ &= \deg(v)f(v) - \sum_{u \in N(v)} f(u) \\ &= (D - A)f(v) \end{aligned}$$

which shows that the graph Laplacian Δ is the degree operator D minus the adjacency operator A , namely

$$\Delta = d^*d = D - A$$

The mapping d is also self-adjoint so for any $f \in \ell^2(V)$,

$$\langle f, \Delta f \rangle_V = \langle f, d^*df \rangle_V = \langle df, df \rangle_E$$

Since we are considering the regular graphs of degree k , we can normalize this graph Laplacian by $\frac{1}{k}$ to $\Delta' := \frac{1}{k}\Delta$. This operator analogues to the Laplace-Beltrami operator on Riemannian manifolds. For regular graphs of degree k , the degree operator is just a k -multiple of the identity operator, i.e. $D = k \cdot \text{id}$. So,

$$\Delta' = \frac{1}{k}\Delta = \text{id} - \frac{1}{k}A$$

where the last operator $\frac{1}{k}A$ actually behaves as an averaging operator on functions $f \in \ell^2(V)$.

The spectrum of eigenvalues for A , is the sequence $k = \lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_n \geq -k$. After averaging, this sequence is in $[-1, 1]$. So the eigenvalues for the normalized Laplacian are $\mu_i := 1 - \frac{\lambda_i}{k}$, forming a sequence

$$0 = \mu_1 \leq \mu_2 \leq \dots \leq \mu_n$$

So now we just finish the settings of the eigenvalues of Laplacian operators. These eigenvalues are used to bound the Cheeger constant. The upper bounds were shown in [Dod84], [Alo86] and [DK86].

Proposition 1.4: Cheeger's inequality

Let Γ be a connected k -regular graph. Then,

$$\frac{\mu_2}{2} \leq \frac{1}{k}h(\Gamma) \leq \sqrt{2\mu_2}$$

Remark In the context of λ_i rather than μ_i , this inequality is

$$\frac{1}{2}(k - \lambda_2) \leq h(\Gamma) \leq \sqrt{2k(k - \lambda_2)}$$

Proof: The 'harder' part $h(\Gamma) \leq \sqrt{2k(k - \lambda_2)}$:

Suppose g is the normalized ($\|g\| = 1$) eigenfunction of the Laplacian operator Δ corresponding to the eigenvalue $\lambda := k - \lambda_2$. From the property of the Laplacian operator, we have the following

$$\langle dg, dg \rangle = \langle g, \Delta g \rangle = \lambda \langle g, g \rangle = \lambda$$

Now let V^+ be the set of all V such that $g(v) > 0$, i.e. $V^+ = \{v \in V : g(v) > 0\}$. Let f be

$$f(v) := \begin{cases} g(v) & v \in V^+ \\ 0 & v \notin V^+ \end{cases}$$

Then we have

$$\begin{aligned} \langle df, df \rangle &= \sum_{v \in V} \deg(v) f^2(v) - \sum_{v \in V} \sum_{u \in V} \delta_{uv} f(v) f(u) \\ &= \sum_{v \in V} f(v) \sum_{u \in V} \delta_{vu} (f(v) - f(u)) \\ &= \sum_{v \in V^+} g(v) \sum_{u \in V} \delta_{vu} (g(v) - f(u)) \\ &= \sum_{v \in V^+} g(v) \left(\sum_{u \in V^+} \delta_{vu} (g(v) - g(u)) + \sum_{u \notin V^+} \delta_{vu} (g(v) - 0) \right) \\ &= \sum_{v \in V^+} g(v) \sum_{u \in V^+} \delta_{vu} (g(v) - g(u)) + \sum_{v \in V^+} g(v)^2 \sum_{u \notin V^+} \delta_{vu} \end{aligned} \quad \deg(v) = \sum_{u \in V} \delta_{uv}$$

Since $g(u) < 0$ for $u \notin V^+$ and $-\sum_{v \in V^+} g(v) \sum_{u \notin V^+} \delta_{vu} g(u) > 0$, then

$$\begin{aligned} \langle df, df \rangle &\leq \sum_{v \in V^+} g(v) \sum_{u \in V} \delta_{vu} (g(v) - g(u)) = \sum_{v \in V^+} g(v) (\Delta g)(v) \\ &= \lambda \sum_{v \in V^+} g(v)^2 = \lambda \langle f, f \rangle \end{aligned} \quad (1)$$

The expression $A := \sum_{e \in E} |f(e^+)^2 - f(e^-)^2|$ can be rewritten as

$$\begin{aligned} \sum_{e \in E} |f(e^+)^2 - f(e^-)^2| &= \sum_{e \in E} |f(e^+) + f(e^-)| |f(e^+) - f(e^-)| \\ &\leq \left(\sum_{e \in E} |f(e^+) + f(e^-)|^2 \right)^{\frac{1}{2}} \left(\sum_{e \in E} |f(e^+) - f(e^-)|^2 \right)^{\frac{1}{2}} \quad \text{Cauchy-Schwartz} \\ &\leq \left(2 \sum_{e \in E} (f(e^+)^2 + f(e^-)^2) \right)^{\frac{1}{2}} \langle df, df \rangle^{\frac{1}{2}} \end{aligned} \quad (2)$$

Let $I(e, v)$ Be an indicator function between an edge e and a vertex v ,

$$I(e, v) := \begin{cases} 1 & e \text{ is incident to } v \\ 0 & \text{else} \end{cases}$$

So, $\sum_{e \in E} f(e^+)^2 + f(e^-)^2$ can be rephrased as

$$\sum_{e \in E} f(e^+)^2 + f(e^-)^2 = \sum_{e \in E} \sum_{v \in V} I(e, v) f(v)^2 = \sum_{v \in V} \deg(v) f(v)^2 \quad (3)$$

Since Γ is a k -regular graph, (3) is exactly $k \langle f, f \rangle$. Combining (1), (2) and (3), we have

$$A \leq \sqrt{2k} \langle f, f \rangle^{\frac{1}{2}} \langle df, df \rangle^{\frac{1}{2}} \leq \sqrt{2k\lambda} \langle f, f \rangle \quad (\dagger)$$

A can be estimated from below by defining level sets as follows: f has finitely many values because f is defined on a finite graph. Arrange those values with 0 in a sequence $\beta_0 = 0 < \beta_1 < \dots < \beta_r$. For each β_i , the level set L_i is

$$L_i := \{v \in V : f(x) \geq \beta_i\}$$

Naturally, $L_r \subseteq \dots \subseteq L_1 \subseteq L_0$. Set $T(i, e)$ to be the indicator function

$$T(i, e) = \begin{cases} 1 & e \in \partial L_i \\ 0 & e \notin \partial L_i \end{cases}$$

Then, A can be rewritten as

$$\begin{aligned}
A &= \sum_{e \in E} |f(e^+)^2 - f(e^-)^2| = \sum_{e \in E} \sum_{i=1}^r T(i, e)(\beta_i^2 - \beta_{i-1}^2) \\
&= \sum_{i=1}^r \sum_{e \in E} T(i, e)(\beta_i^2 - \beta_{i-1}^2) \\
&= \sum_{i=1}^r |\partial L_i|(\beta_i^2 - \beta_{i-1}^2) \qquad \sum_{e \in E} T(i, e) = |\partial L_i|
\end{aligned}$$

By the definition of Cheeger constant, $|\partial L_i| \geq h(\Gamma)|L_i|$ for all L_i . Then,

$$\begin{aligned}
A &= \sum_{i=1}^r |\partial L_i|(\beta_i^2 - \beta_{i-1}^2) \geq h(\Gamma) \sum_{i=1}^r |L_i|(\beta_i^2 - \beta_{i-1}^2) \\
&= h(\Gamma) \left(|L_r|\beta_r^2 + \sum_{i=1}^{r-1} \beta_i^2(|L_i| - |L_{i+1}|) \right) \quad (\dagger)
\end{aligned}$$

Notice that for each $1 \leq i < r$, $f(x) = \beta_i \Leftrightarrow x \in L_i \setminus L_{i+1}$. $|L_i| - |L_{i+1}| = |\{v \in V : f(x) = \beta_i\}|$. For each $1 \leq i < r$,

$$\beta_i^2(|L_i| - |L_{i+1}|) = \sum_{v \in V} \mathbf{1}_{f^{-1}(\{\beta_i\})}(v) f(v)^2$$

Especially for $i = r$,

$$\beta_r^2 |L_r| = \sum_{v \in V} \mathbf{1}_{f^{-1}(\{\beta_r\})}(v) f(v)^2$$

So, $(\dagger)$ has a simpler form

$$h(\Gamma) \sum_{i=1}^r \sum_{v \in f^{-1}(\{\beta_i\})} f(v)^2 = h(\Gamma) \sum_{v \in V} f(v)^2 = h(\Gamma) \langle f, f \rangle$$

From an upper bound $(\dagger)$ and a lower bound $(\ddagger)$ of A ,

$$h(\Gamma) \leq \sqrt{2k\lambda}$$

The 'easier' part $\frac{1}{2}(k - \lambda_2) \leq h(\Gamma)$:

Let A, B be an arbitrary partition of V , namely $A \sqcup B = V$. Set $|A| = a$ and $|B| = b$. Without loss of generality assume $a \leq b$. Defining a function f to be

$$f(x) := \begin{cases} b & x \in A \\ -a & x \in B \end{cases}$$

Then this function is in $\ell_0^2(V)$. Notice that from (1), we have

$$\lambda \leq \frac{\|df\|}{\|f\|} \stackrel{\dagger\dagger}{=} \frac{n^2 |E(A, B)|}{nab} \stackrel{\ddagger\dagger}{\leq} \frac{|E(A, B)|}{2a}$$

$\dagger\dagger$ is just to apply the equality $\langle f, f \rangle = \sum_{v \in V} f(v) \sum_{u \in V} \delta_{uv}(f(v) - f(u))$ and summing δ_{uv} over $v \in A$ and $u \in B$ ($\sum_{v \in A} \sum_{u \in B} \delta_{uv} = |E(A, B)|$). While, $\langle f, f \rangle = \sum_{v \in A} f(v)^2 + \sum_{v \in B} f(v)^2 = ab^2 + ba^2 = nab$.

$\ddagger\dagger$ Because $a \leq b$, then $b \geq \frac{1}{2}n$.

Since $|A| \leq |B|$ is assumed. Whatever A and B are, it is always the case that $|A| = \min\{|A|, |B|\}$. Since this inequality works for arbitrary partitions, then

$$\lambda \leq \frac{1}{2} \min \frac{|E(A, B)|}{\min\{|A|, |B|\}} = h(\Gamma)$$

□

2 Constructing expander graphs

2.1 Cayley graphs

Definition 2.1: Cayley graphs

Let G be a group and $S \subseteq G$ be a generating set of G . The Cayley graph of G under S , denoted $\Gamma(G, S)$, is a graph whose
 vertex set is G ;
 edge set is $\{(sg, g) | g \in G, s \in S\}$.

Remark If the graph is undirected, S should be symmetric !!!

Proposition 2.1: Characterisation of expansion properties of Cayley graphs

Let $\Gamma(G, S)$ be a finite Cayley graph. For any given ε ,
 (1) $\Gamma(G, S)$ is a one-sided ε -Cayley graph $\Leftrightarrow S$ generates G .
 (2) $\Gamma(G, S)$ is a two-sided ε -Cayley graph $\Leftrightarrow S$ generates G and S intersects each index 2 subgroup, i.e. subgroup $H \leq G$ with $[G : H] = 2$.

Proof:

□

2.2 Schreier graphs

Definition 2.2: Schreier Graph

Let G be a group, and let $H \leq G$ be a subgroup. Let $S \subset G$ be a finite generating set of G , which is typically assumed to be symmetric (i.e., $s \in S \implies s^{-1} \in S$) and not containing the identity ($1 \notin S$). The Schreier graph $\text{Sch}(G, H, S)$ is a graph defined as follows: Vertex Set (V):
 The vertices are the right cosets of H in G . Thus, $V = G/H = \{Hg \mid g \in G\}$.
 Edge Set (E): Two vertices Hg_1 and Hg_2 are connected by a directed edge labeled $s \in S$ if and only if:

$$Hg_1 s = Hg_2$$

Because S is symmetric, every directed edge (Hg, Hgs) has a corresponding reverse edge $(Hgs, Hgss^{-1}) = (Hgs, Hg)$, allowing us to treat the graph as undirected.

Remark If we choose the trivial subgroup $H = \{1\}$, the cosets Hg are just the elements g themselves, and the Schreier graph $\text{Sch}(G, \{1\}, S)$ collapses exactly into the Cayley graph $\text{Cay}(G, S)$. Therefore, the Cayley graph is simply the **universal covering space** of the Schreier graph, and the Schreier graph is formed by taking the quotient of the Cayley graph by the action of the subgroup H .

Property of groups of Lie type (e.g. $\text{SL}_2(\mathbb{Z})$)

2.3 Kazhdan's property (T)

First, we will see that the Kazhdan's property (T) associates to the expansion of Cayley graphs. It was first collected and introduced by Kazhdan in 1967 and was originally used to prove some discrete subgroups are finitely generated. However, it was Margulis to construct the first expanding Cayley graphs by this property (T). But, expansions of Cayley graphs do not always satisfy the property (T). We would create some Cayley graphs without using property (T) afterwards.

Definition 2.3: Kazhdan's property (T)

Let H be a (complex) Hilbert space. Let G be a σ -compact, locally compact group and $\rho : G \rightarrow U(H)$ be a unitary representation. Then, for any given $\varepsilon > 0$ and compact subgroup K of G , a unit vector $u \in H$ is an (ε, K) -invariant vector $\Leftrightarrow \forall g \in K, \|\rho(g)(u) - u\| < \varepsilon$.
 G is said to have the **Kazhdan's property (T)** $\Leftrightarrow \exists \varepsilon > 0$ and a compact subset $K \subseteq G$ such that every unitary representation ρ of G that has an (ε, K) -invariant unit vector, has a nonzero invariant vector.

Remark The core logic of this definition is a strict conditional mechanism. It states that there exist a global compact set K and an error tolerance $\epsilon > 0$ such that: If any unitary representation contains a unit vector that is "almost invariant" under the action of K (i.e., satisfying the (ϵ, K) -invariant condition), Then this representation space must contain a strictly non-zero, absolutely invariant vector. In other words, this property tolerates no "pseudo-invariance." A representation either lacks any trace of invariance, or it must solidly provide a genuinely invariant vector.

In standard analysis, or in groups lacking Property (T), we expect a continuum: one can always find arbitrarily small quantities and vectors that infinitely approximate invariance without ever truly achieving it. However, Property (T) dictates that the group structure simply does not tolerate these infinitesimal gaps.

If the variation of a vector falls below a certain definitive threshold ϵ , the continuous approximation abruptly stops, and the vector automatically snaps into perfect alignment. It transitions instantly from being "almost identical" to being strictly invariant. In other words, any gap smaller than ϵ simply vanishes. The space between "sufficiently close" and "absolutely perfect" is completely hollowed out, leaving no room for intermediate, almost-invariant states.

Example For $d \geq 3$, $\mathrm{SL}_d(\mathbb{Z})$ and $\mathrm{SL}_d(\mathbb{R})$ have property (T). For $d < 3$, they do not have property (T) anymore. Free non-abelian groups F_k does not have property (T), and every finitely generated infinite abelian group does not have property (T).

Corollary 2.0.1 (Property (T) and amenability)

Corollary 2.0.2 *If a group G is amenable and has Kazhdan property (T), then G is compact.*

Proof:

□

Corollary 2.0.3 (property of non-abelian free groups) *If G is a non-abelian free group, then G has neither amenability nor Kazhdan property (T).*

Corollary 2.0.4 (property between groups and its lattices) *Let G be a group.*

Corollary 2.0.5 $(\mathrm{SL}_2(\mathbb{R}))$ $\mathrm{SL}_2(\mathbb{R})$ *does not have Kazhdan property (T) nor amenability.*

Proof: $\mathrm{SL}_2(\mathbb{R})$ has a lattice $\mathrm{SL}_2(\mathbb{Z})$. By pingpong lemma, $\mathrm{SL}_2(\mathbb{Z})$ has a free subgroup F (in matrix notation, generated by $\begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix}$ and $\begin{pmatrix} 1 & 0 \\ 2 & 1 \end{pmatrix}$) of finite index. This non-abelian group F is another lattice of $\mathrm{SL}_2(\mathbb{R})$. By 2.0.3, F is not amenable nor has Kazhdan property (T). By 2.0.4, $\mathrm{SL}_2(\mathbb{R})$ is not amenable nor has Kazhdan property (T). □

To understand the behavior of $\mathrm{SL}_d(\mathbb{R})$ where $d \geq 3$, the most straightforward way is to study the case when $d = 3$. This $d = 3$ case will inspire us the most general case. So, the first question is how to recover $\mathrm{SL}_3(\mathbb{R})$ from $\mathrm{SL}_2(\mathbb{R})$ or how to decompose $\mathrm{SL}_3(\mathbb{R})$ into $\mathrm{SL}_2(\mathbb{R})$ and any other components. This is

Theorem 2.1 (Structure of $\mathrm{SL}_3(\mathbb{R})$) $\mathrm{SL}_3(\mathbb{R}) \cong \mathrm{SL}_2(\mathbb{R}) \ltimes \mathbb{R}^2$

2.4 Zigzag product constructions

3 Applications of expander graphs:

4 Applications of expander graphs: to certain sieving problems in analytic number theory

References

- [Dod84] Jozef Dodziuk. “Difference equations, isoperimetric inequality and transience of certain random walks”. In: *Transactions of the American Mathematical Society* 284.2 (1984), pp. 787–794.
- [Alo86] Noga Alon. “Eigenvalues and expanders”. In: *Combinatorica* 6.2 (1986), pp. 83–96.
- [DK86] Jozef Dodziuk and Wilfrid S Kendall. “Combinatorial Laplacians and isoperimetric inequality”. In: *From local times to global geometry, control and physics*. Ed. by K. D. Elworthy. Vol. 150. Pitman Research Notes in Mathematics Series. Harlow: Longman Scientific & Technical, 1986, pp. 68–74.
- [Big93] N. Biggs. *Algebraic Graph Theory*. Cambridge Mathematical Library. Cambridge University Press, 1993. ISBN: 9780521458979. URL: <https://books.google.ca/books?id=6TasRmIFOxQC>.